

### ELECTIVE-III

| UEC643: AUDIO AND VIDEO PROCESSING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |    |
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| <p><b>Course Objective:</b> To provide students with the knowledge of basic characteristics of speech and video signal. To describe basic algorithms of speech analysis common to many applications. To make students acquainted with state-of-the-art video processing techniques, their technical details and challenges.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |    |
| <p>Syllabus</p> <p><b>Part A</b></p> <p><b>Introduction:</b> Introduction to speech processing, Characteristics of speech signals, Digitization of speech signal.</p> <p><b>Digital Models for Speech signals:</b> Human Speech production, Acoustic Phonetics. Uniform Tube Modeling of Speech Production.</p> <p><b>Speech Recognition:</b> Automatic speech recognition system, architecture, features extraction, effect of windowing, spectrogram, cepstral analysis, Hidden Markov Models (HMM), Viterbi algorithm, Vector Quantization (VQ).</p> <p><b>Speaker recognition:</b> Speaker verification/authentication vs. speaker identification, Automatic Speaker Verification system, pattern matching (e.g. dynamic time warping (DTW), VQ, HMM).</p> <p><b>Part B</b></p> <p><b>Introduction:</b> Video formats, Capturing of video signals, Color space, Quality</p> <p><b>Video Motion Estimation:</b> general methodologies-Block matching algorithm, Deformable blocks matching algorithm, Region based motion estimation, Multi-resolution motion estimation.</p> <p><b>Video Coding:</b> Transform and Quantization, Content dependent video coding, Region based video coding, Object based video coding, Knowledge based video coding, Semantic video coding, Scalable Video Coding.</p> <p><b>Video Compression:</b> inter-frame and intra-frame compression, Lossy and Loss less compression techniques, 3D video compression standard (H.264).</p> <p><b>Familiarization with standards:</b> IEEE 1857-2013, IEEE 1857.5-2015, IEEE H.264/AVC video coding standard.</p> |   |   |   |    |
| <p><b>Laboratory Work:</b> Students have to write MATLAB /Python programs to gain basic understanding of speech signal, to apply the fundamental concepts of signal processing on speech signals, extraction of fundamental frequency, dividing raw video into frames, compression, coding and reframing the video.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |    |
| <p><b>Course Learning Objectives (CLO)</b></p> <p>The students will be able to:</p> <ol style="list-style-type: none"> <li>1. Model speech signals and represent them digitally.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |

2. Analyze speech and speaker recognition systems.
3. Explain video formats, color spaces and compression techniques.
4. Implement algorithms for video motion estimation and coding.

**Text Books**

1. Rabiner, L. and Schafer, R., *Digital Processing of Speech Signals*. Signal Processing, Prentice-Hall (1978).
2. Deller, J., Proakis, J. and Hansen, J., *Discrete-Time Processing of Speech Signals*, IEEE(1993).
3. M. Tekalp, *Digital Video Processing*, Prentice Hall, 2nd Edition, 2015.
4. Iain E. Richardson, *THE H.264 ADVANCED VIDEO COMPRESSION STANDARD*, John Wiley and Sons, Ltd., 2003.

**Reference Books**

1. Borden, G. and Harris, K., *Speech Science Primer*, 2nd Edition, Williams and Wilkins,(2006).
2. Furui, S., *Digital Speech Processing, Synthesis and Recognition*, CRC (2001).
3. Alan C. Bovik, *The Essential Guide to Video Processing*, Academic Press; 2009
4. J. W. Woods, *Multidimensional Signal, Image, and Video Processing and Coding*, Academic Press, 2011.